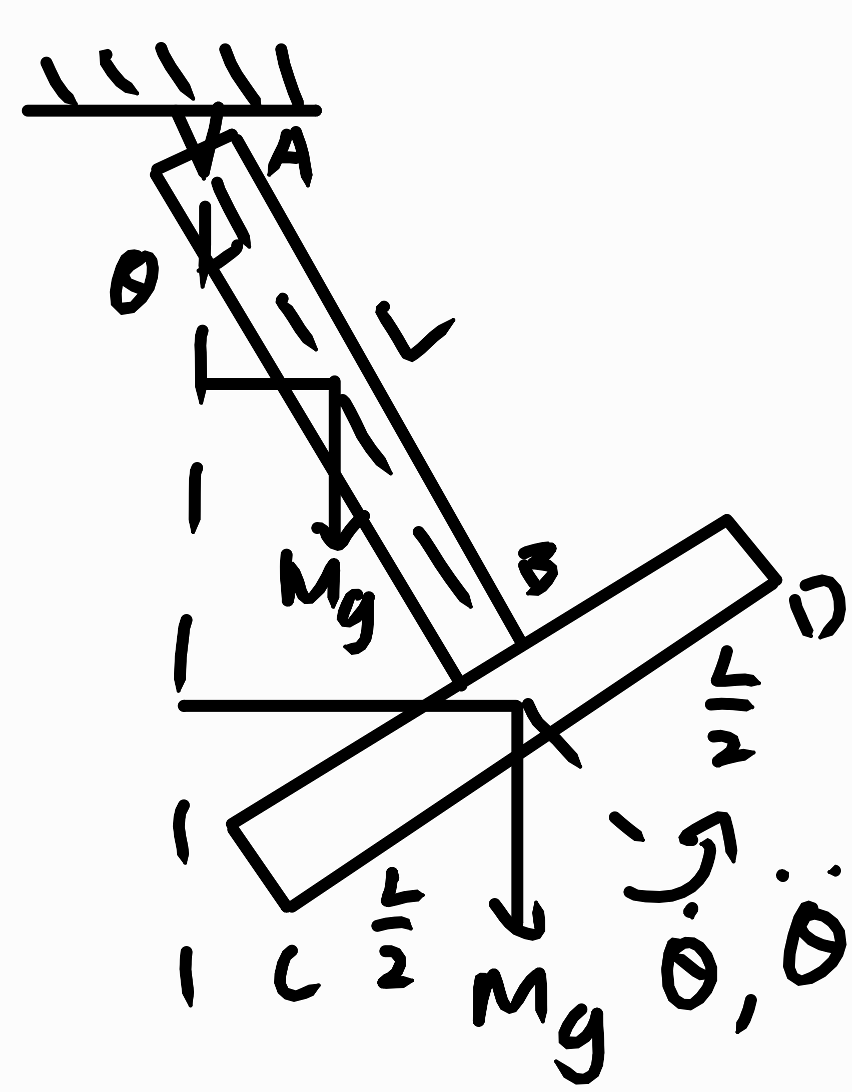
1) From Newton's 2^{nd} Law



$$J_A \ddot{\theta} = \sum M_A \quad -(1)$$

$$J_A = J_{AB@A} + J_{CD@A}$$
$$= \frac{1}{3}ML^2 + \left(J_{CD@A} + M(AB)^2\right)$$
$$= \frac{1}{3}ML^2 + \left(\frac{1}{12}ML^2 + ML^2\right)$$
$$= \frac{17}{12}ML^2$$

$$\sum M_A = M_{AB@A} + M_{CD@A}$$
$$= -Mg\frac{L}{2}\sin\theta - MgL\sin\theta$$
$$= -\frac{3}{2}MgL\sin\theta$$
$$\approx -\frac{3}{2}MgL\theta \quad \because \sin\theta \approx \theta \text{ when } \theta \text{ is small}$$

Substituting into equation (1):

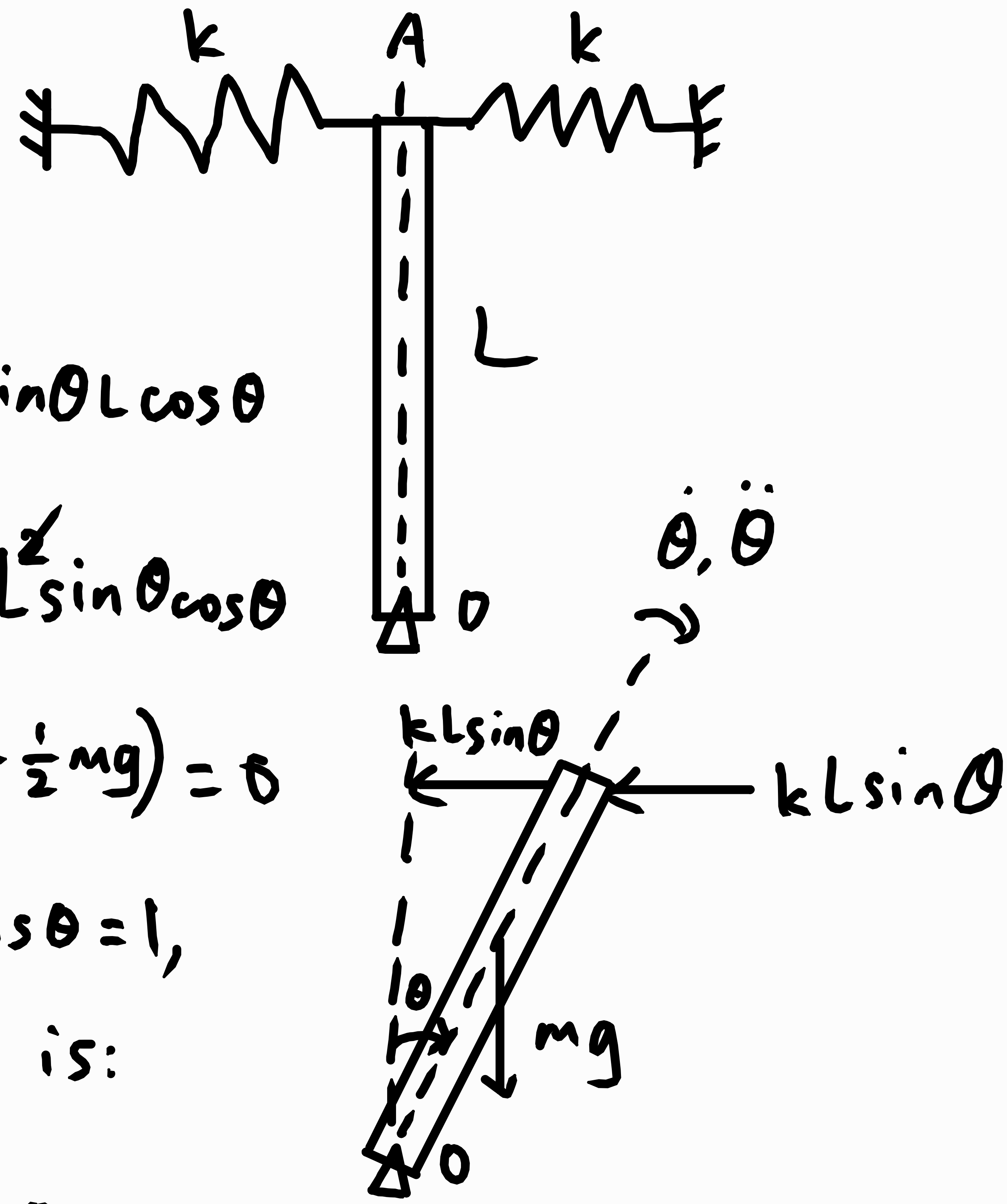
$$\frac{17}{12}ML^2\ddot{\theta} = -\frac{3}{2}MgL\theta$$

$$\underbrace{\left(\frac{17}{12}ML^2\right)}_{\text{Effective inertia}}\ddot{\theta} + \underbrace{\left(\frac{3}{2}MgL\right)}_{\text{Effective stiffness}}\theta = 0$$

Natural frequency:

$$f_n = \frac{1}{2\pi}\sqrt{\frac{\text{Effective stiffness}}{\text{Effective inertia}}} = \frac{1}{2\pi}\sqrt{\frac{\frac{3}{2}\cancel{M}g\cancel{L}}{\frac{17}{12}\cancel{M}L^{\cancel{2}}}} = \frac{1}{2\pi}\sqrt{\frac{18g}{17L}} \text{ Hz (shown)}$$

2) $J_o \ddot{\theta} = \sum M_o$



$$J_o = \frac{1}{3} mL^2$$

$$\sum M_o = mg\frac{L}{2}\sin\theta - 2kL\sin\theta L\cos\theta$$

$$\therefore \frac{1}{3} m\cancel{L^2}\ddot{\theta} = mg\frac{\cancel{L}}{2}\sin\theta - 2k\cancel{L^2}\sin\theta\cos\theta$$

$$\left(\frac{1}{3}mL\right)\ddot{\theta} + \sin\theta\left(2kL\cos\theta - \frac{1}{2}mg\right) = 0$$

Since $\sin\theta \approx \theta$, $\cos\theta = 1$,

the equation of motion is:

$$\left(\frac{1}{3}mL\right)\ddot{\theta} + \left(2kL - \frac{1}{2}mg\right)\theta = 0$$

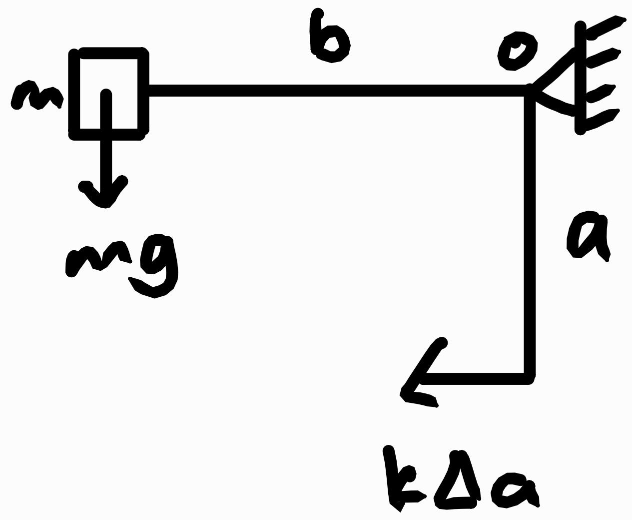
$$f_n = \frac{1}{2\pi}\sqrt{\frac{\text{Effective stiffness}}{\text{Effective inertia}}}$$

$$= \frac{1}{2\pi}\sqrt{\frac{2kL - \frac{1}{2}mg}{\frac{1}{3}mL}}$$

$$= \frac{1}{2\pi}\sqrt{\frac{6k}{m} - \frac{3g}{2L}} \text{ Hz}$$

3) $\sum M_0 = 0$

Static analysis



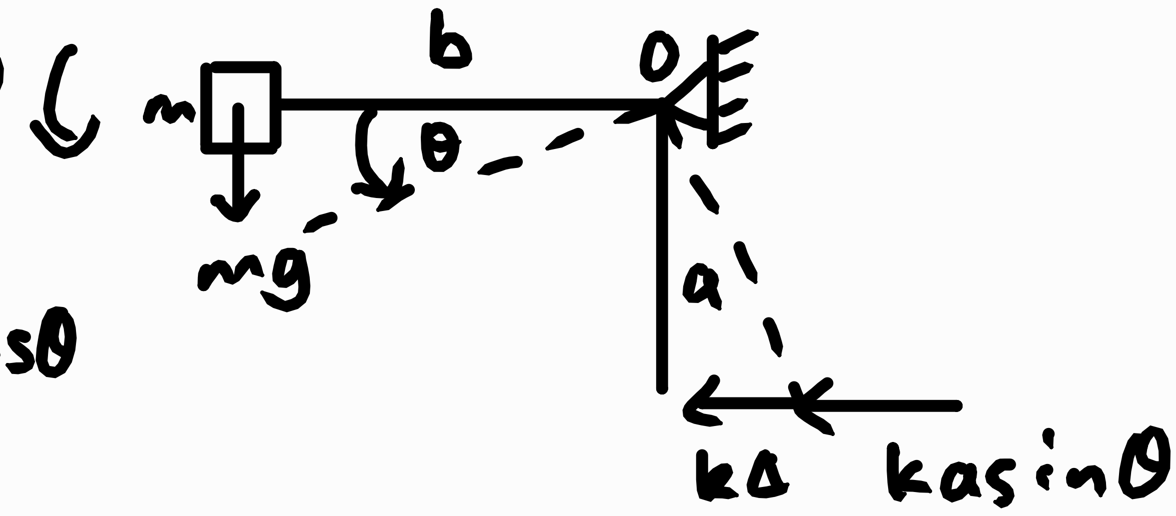
$mgb = k\Delta$

$mgb - k\Delta a = 0$

Dynamic analysis

$J_0 \ddot{\theta} = \sum M_0$

$\dot{\theta}, \ddot{\theta}$



$mb^2\ddot{\theta} = mgb\cos\theta - k\Delta a\cos\theta$
$\quad - ka\sin\theta\, a\cos\theta$

$mb^2\ddot{\theta} = mgb\cos\theta - ka\cos\theta(\Delta + a\sin\theta)$

Using small angle approximation:

$$\sin\theta \simeq \theta \ , \quad \cos\theta \simeq 1$$

$\therefore mb^2\ddot{\theta} = mgb - ka(\Delta + a\theta)$

$mb^2\ddot{\theta} = mgb - ka\Delta - ka^2\theta$

From the static analysis,

$$mgb - ka\Delta = 0$$

$\therefore mb^2\ddot{\theta} = 0 - ka^2\theta$

$mb^2\ddot{\theta} + ka^2\theta = 0$

$$f_n = \frac{1}{2\pi}\sqrt{\frac{\text{Effective stiffness}}{\text{Effective inertia}}}$$

$$= \frac{1}{2\pi}\sqrt{\frac{ka^2}{mb^2}} = \frac{1}{2\pi}\frac{a}{b}\sqrt{\frac{k}{m}} \text{ Hz}$$

4) $\Delta = \dfrac{FL^3}{3EI}$ where $I = \dfrac{wt^3}{12}$

stiffness of the cantilever, $k_b = \dfrac{F}{\Delta}$

$$= \frac{\cancel{F}}{\frac{\cancel{F}L^3}{3EI}}$$

$$= \frac{3EI}{L^3}$$

$$= \frac{3Ewt^3}{12L^3} \quad \because I = \frac{wt^3}{12}$$

For springs in series:

$$\frac{1}{k_{eff}} = \frac{1}{k} + \frac{1}{k_b}$$

$$\frac{1}{k_{eff}} = \frac{k_b + k}{k k_b}$$

$$k_{eff} = \frac{k k_b}{k_b + k}$$

$$f_n = \frac{1}{2\pi}\sqrt{\frac{\text{Effective stiffness}}{\text{Effective mass}}} = \frac{1}{2\pi}\sqrt{\frac{k_{eff}}{m}}$$

$$= \frac{1}{2\pi}\sqrt{\frac{k k_b}{m(k_b + k)}}$$

5) $f_{n,m=0} = \frac{1}{0.4} = 2.5\,Hz$

$f_{n,m=30} = \frac{1}{0.5} = 2.0\,Hz$

$m = 0$ $\qquad\qquad$ $m = 30$

$m_{eff}\ddot{x} + k_{eff}x = 0$

$f_n = \frac{1}{2\pi}\sqrt{\frac{k_{eff}}{m_{eff}}}$

$(2\pi f_n)^2 = \frac{k_{eff}}{m_{eff}}$

$(2\pi(2.5))^2 = \frac{k_{eff}}{m_{eff}}$

$k_{eff} - 25\pi^2 m_{eff} = 0 \quad -(1)$

$(m_{eff}+30)\ddot{x} + k_{eff}x = 0$

$f_n = \frac{1}{2\pi}\sqrt{\frac{k_{eff}}{m_{eff}+30}}$

$(2\pi f_n)^2 = \frac{k_{eff}}{m_{eff}+30}$

$(2\pi(2.0))^2(m_{eff}+30) = k_{eff}$

$k_{eff} - 16\pi^2 m_{eff} = 480\pi^2 \quad -(2)$

Solving the equations:

$m_{eff} = \frac{160}{3} \approx 53.3\,kg$

$k_{eff} = 13159.47253 \approx 13159.47\,Nm^{-1}$